

A BASH-BASED LINUX ADMINISTRATION TOOL

Linux Admin Dashboard

```

$ ./dashboard.sh

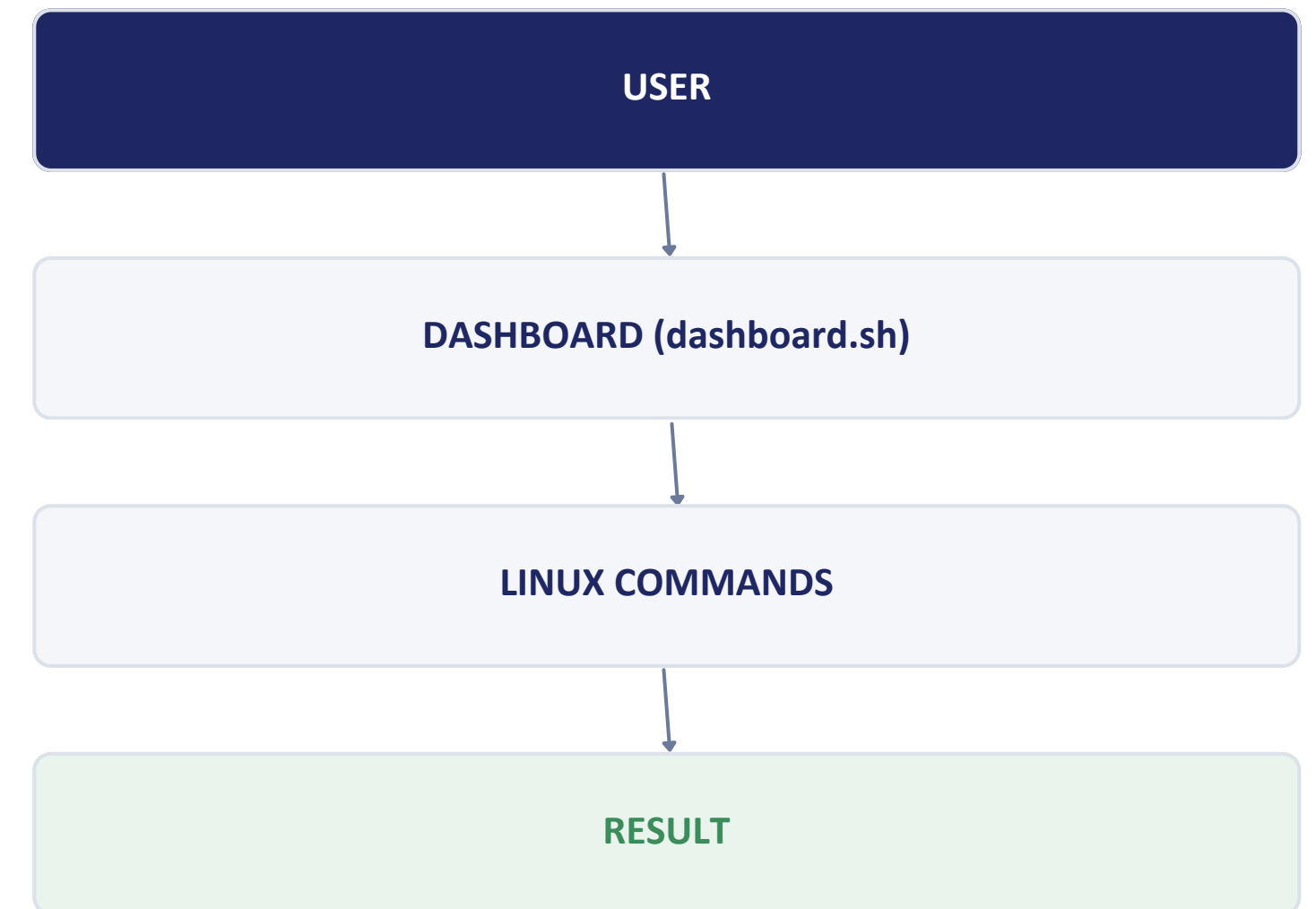
=====
          LINUX ADMIN DASHBOARD
=====

1. System Information
2. Users and Groups
3. File and Directory Management
4. Permission Audit
5. Text and Log Analysis
6. Backup Management
7. Report Generation
8. Exit

Enter your choice: _

```

Eman Ahmad



An interactive Bash menu that routes each choice to real Linux commands, validated at every step.

Main Menu & User Flow

=====

LINUX ADMIN DASHBOARD

=====

1. System Information

2. Users and Groups

3. File and Directory Management

4. Permission Audit

5. Text and Log Analysis

6. Backup Management

7. Report Generation

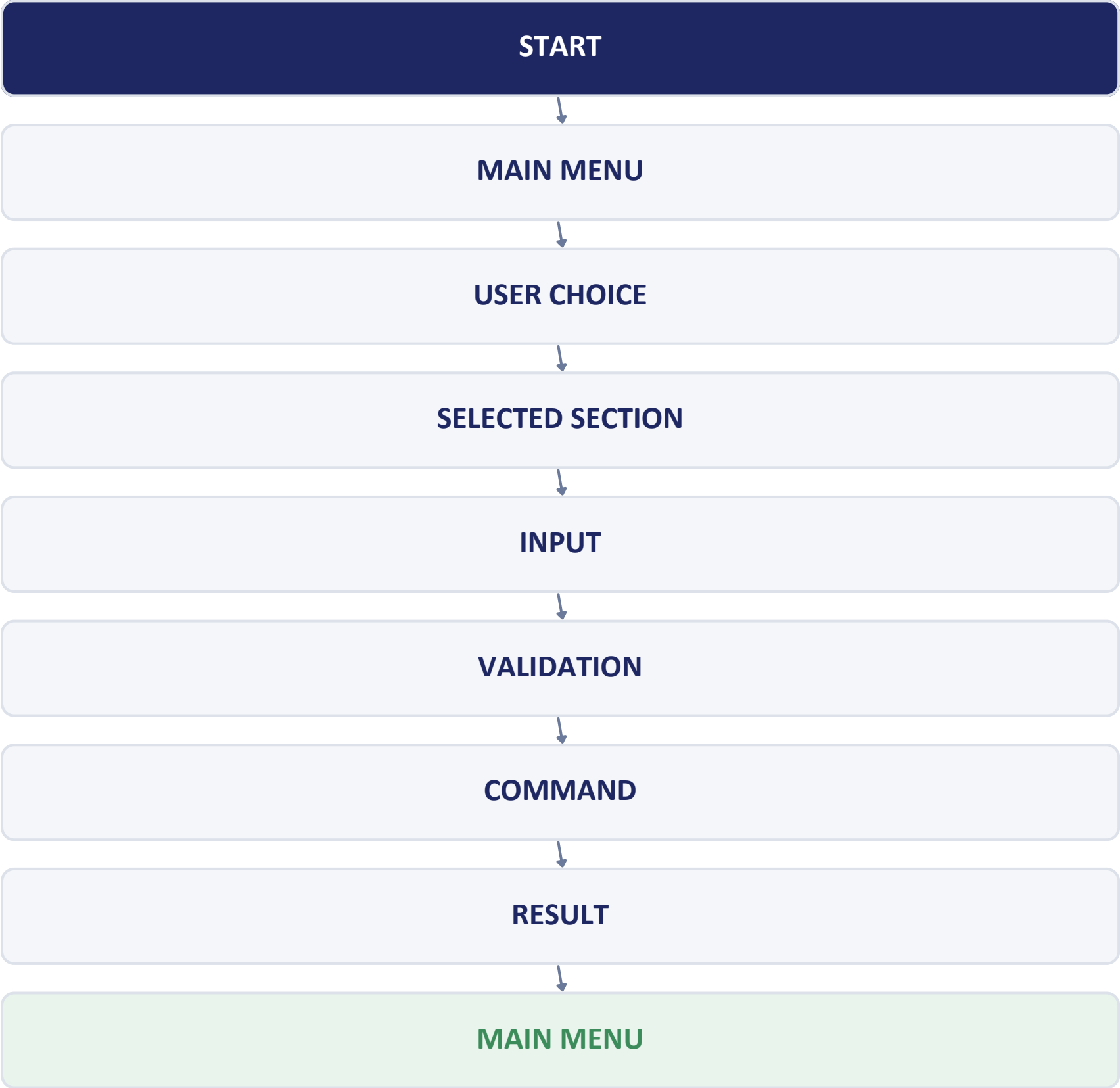
8. Exit

Enter your choice: _

INVALID INPUT CASE

User enters:
9

"Invalid choice."



System Information

```
=====
                        SYSTEM INFORMATION
=====

1. Current User
2. Current Directory
3. Current Date
4. Operating System
5. Kernel Release
6. Machine Architecture
7. Back to Main Menu

Current User:
eman
Operating System:
Linux
Kernel Release:
6.x.x
```

COMMANDS USED

\$

whoami

\$

pwd

\$

date

\$

uname

CASE HANDLED

User chooses:
7

→

Returns to Main Menu

Every section in this dashboard ends with a 'Back to Main Menu' option so the user never gets stuck.

Users & Groups

```
=====
                        USERS AND GROUPS
=====

1. Current User
2. User Groups
3. Add User to Group
4. Remove User from Group
5. Add User
6. Delete User
7. Back to Main Menu
```

CASE 1 — EMPTY USERNAME

Username:
[EMPTY]

→
"Username cannot be empty."

CASE 2 — EMPTY GROUP

Username: eman
Group: [EMPTY]

→
"Username and group cannot be empty."

CASE 3 — VALID INPUT

Username: eman
Group: developers

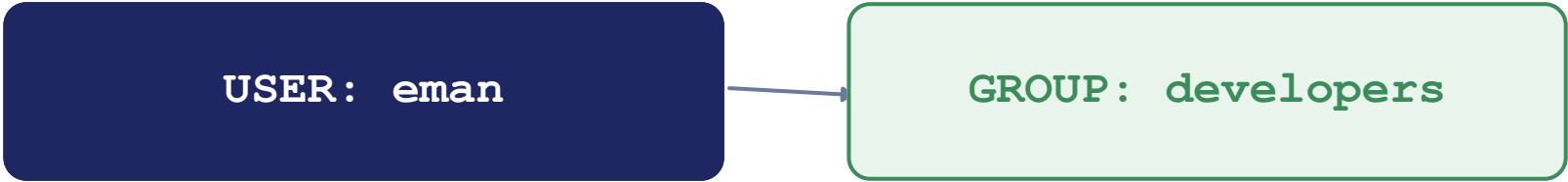
→
Operation executed

CASE 4 — COMMAND FAILS

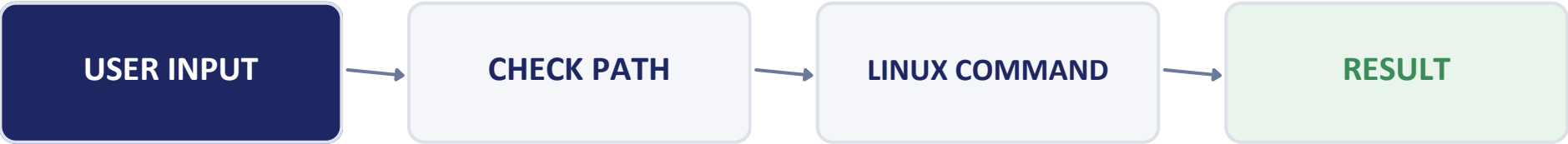
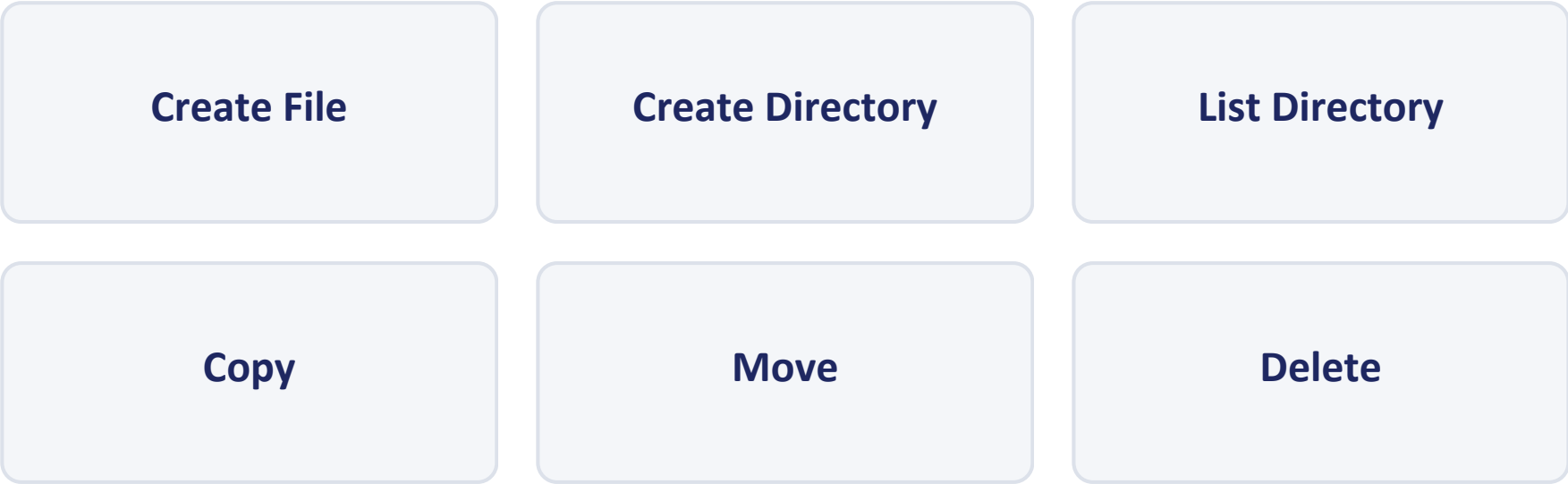
Linux command
returns an error

→
"Failed to perform operation."

USER-GROUP MEMBERSHIP



File & Directory Management



EMPTY PATH

File path:
[EMPTY]

→

"File path cannot be empty."

PATH EXISTS

Directory
already exists

→

"Path already exists."

MISSING SOURCE

Source file
does not exist

→

"Source file does not exist."

VALID INPUT

File path:
notes.txt

→

"File created successfully."

Permission Audit

=====

PERMISSION AUDIT

=====

1. Check File Permission

2. Check Directory Permission

3. Add Execute Permission

4. Remove Execute Permission

5. Check File Owner

6. Check File Details

7. Back to Main Menu

-rwxr-xr--

OWNER	GROUP	OTHERS
rwx	r-x	r--

Owner: eman Group: pc

FILE EXISTS

chmod / ls -l
target file



Display permissions

FILE MISSING

Target file
not found



"File does not exist."

VALID FILE

chmod u+x
script.sh



"Execute permission added."

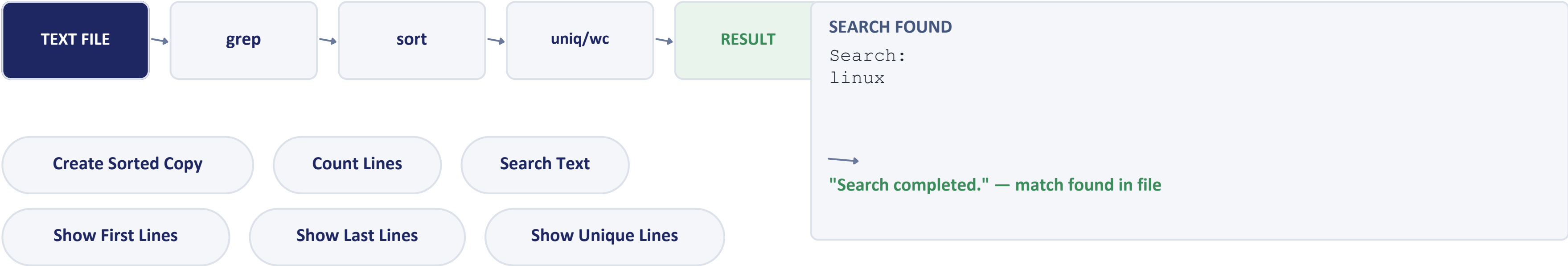
COMMAND FAILS

chmod returns
an error



"Failed to change permission."

Text Analysis



head → shows the first 10 lines
tail → shows the last 10 lines

This section turns raw log files into useful information using a small chain of Linux text-processing tools.

FILE EXISTS
File:
notes.txt

→
Valid file — processing

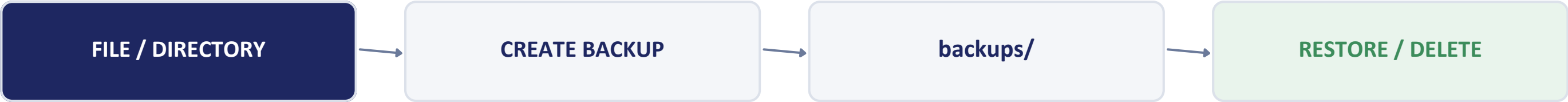
FILE MISSING
File:
missing.txt

→
"File does not exist."

SEARCH NOT FOUND
Search:
python

→
"Text not found."

Backup Management



Create Backup

Show Backup Files

Restore Backup

Delete Backup

VALID SOURCE
Source file
is valid

→
"Backup created successfully."

EMPTY NAME
Backup name:
[EMPTY]

→
"Backup name cannot be empty."

MISSING SOURCE
Source file
does not exist

→
"Source does not exist."

COMMAND FAILS
cp fails during
backup/restore

→
"Backup failed."

Report Generation

SYSTEM REPORT

system_report.txt

USER REPORT

user_report.txt

FILE REPORT

file_report.txt

```
FILE INFORMATION
=====
File:      notes.txt
Details:   -rw-r--r--
Lines:     5
```

VALID FILE

Requested file is valid

→

Report generated.

FILE MISSING

Requested file not found

→

"File does not exist."

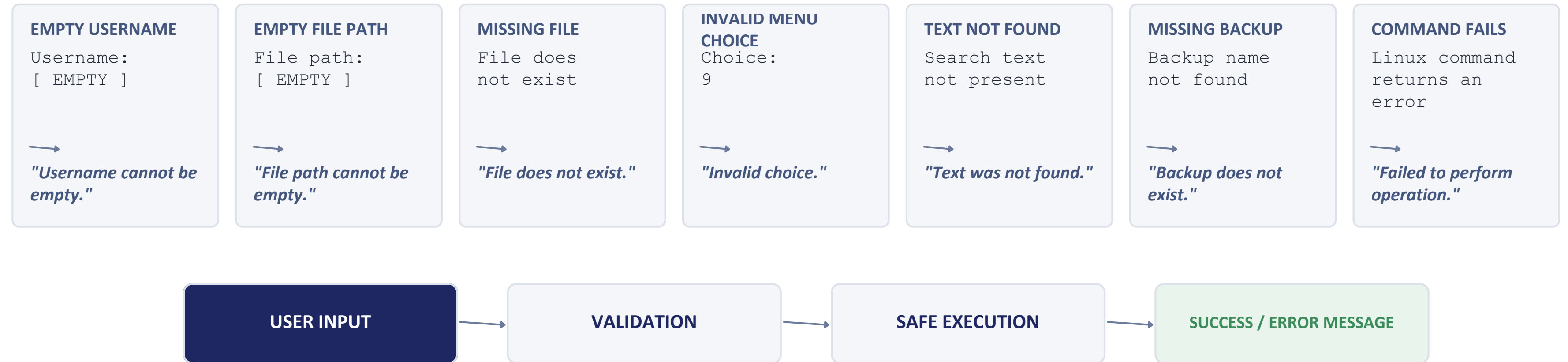
REPORT MISSING

Requested report not found

→

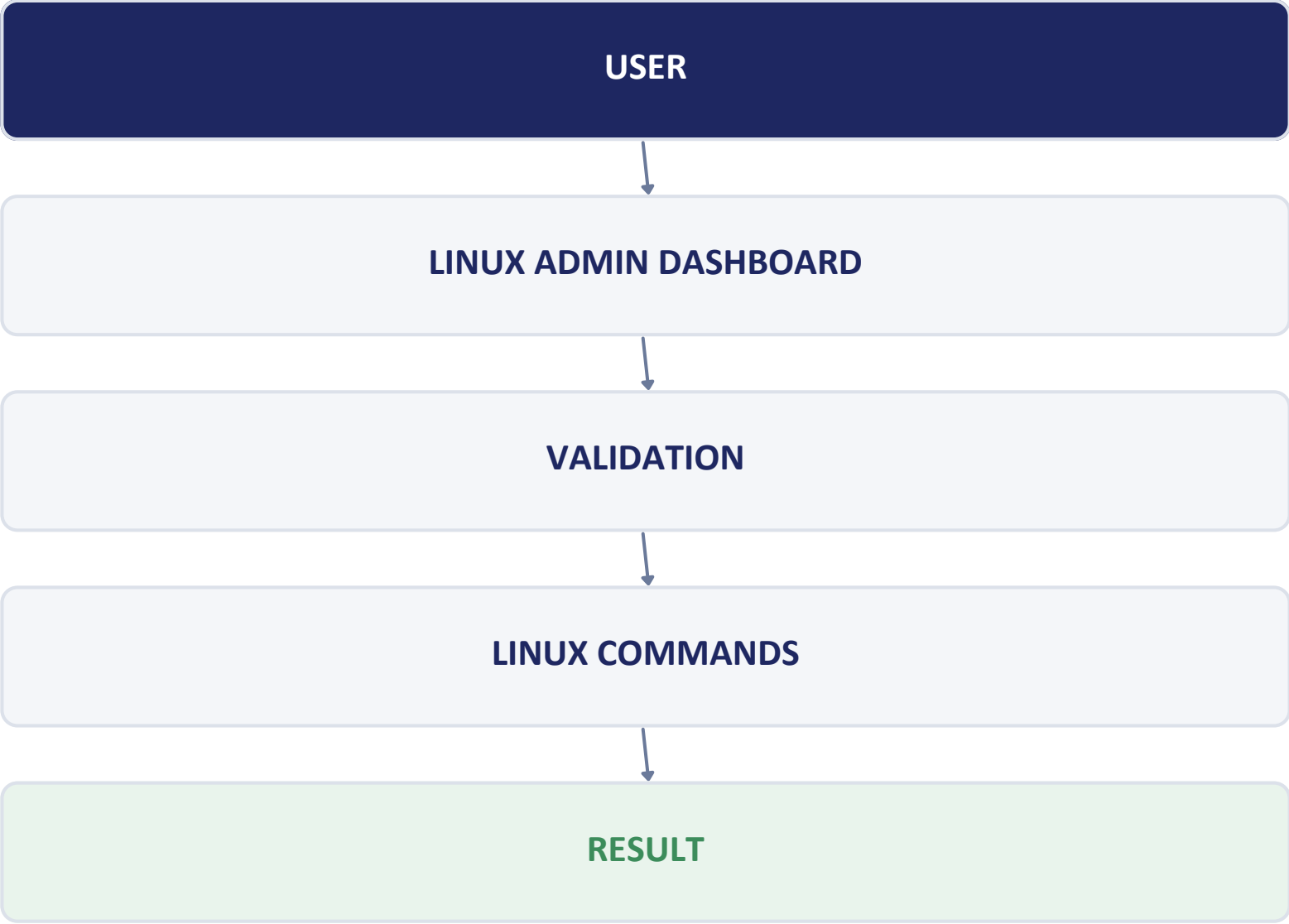
"Report does not exist."

Validation & Error Handling



Validation runs before every command in the dashboard — the program never assumes the user's input is correct.

Project Summary



- ✓ Bash Scripting
- ✓ Linux Administration
- ✓ File Management
- ✓ Permissions
- ✓ Text Processing
- ✓ Backup Management
- ✓ Report Generation
- ✓ Error Handling

“One dashboard for common Linux administration tasks.”

